

Answer Key — Core Mathematics Grade 10

Sub-Strand 2.4: Trigonometry 1 • Lesson 4: The Hidden Link: Discovering $\tan \theta = \sin \theta / \cos \theta$

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	A	B	A	A	C	A

USE AFTER: OBSERVE PHASE — Observe: adding the ' $\sin \theta \div \cos \theta$ ' fourth column and comparing it with the $\tan \theta$ column

Q1. Learners divided $\sin \theta$ by $\cos \theta$ for $\theta = 20^\circ, 30^\circ, 40^\circ, 50^\circ, 60^\circ$ and 70° , and compared each answer with the $\tan \theta$ value in their table. Every row matched, apart from tiny rounding differences. What is the best conclusion from this pattern?

- A. $\sin \theta \div \cos \theta$ is never equal to $\tan \theta$, but the values are close
- B. $\sin \theta \div \cos \theta$ is sometimes equal to $\tan \theta$, depending on the angle
- C. $\sin \theta \div \cos \theta$ is equal to $\tan \theta$ for every acute angle tested, so it looks like an identity**
- D. $\sin \theta \div \cos \theta$ is equal to $\cos \theta \div \sin \theta$ for these angles

Correct answer: C. All six rows agreed, which suggests $\sin \theta \div \cos \theta = \tan \theta$ is always true for acute angles. The lesson then proves this algebraically.

USE AFTER: EXPLAIN PHASE — Explain: teacher's board proof of $\tan \theta = \sin \theta / \cos \theta$

Q2. In a right-angled triangle, $\sin \theta = O/H$ and $\cos \theta = A/H$, where O is the opposite side, A the adjacent side and H the hypotenuse. What is $(O/H) \div (A/H)$ after simplifying?

- A. O/A**
- B. $(O \times A)/H$
- C. A/O
- D. O/H

Correct answer: A. $(O/H) \div (A/H) = (O/H) \times (H/A)$. The H cancels, leaving O/A, which is the definition of $\tan \theta$.

USE AFTER: EXPLAIN PHASE — Explain: Problem C, finding $\tan 50^\circ$ from $\sin 50^\circ$ and $\cos 50^\circ$

Q3. A Kenya Power engineer's calculator has a broken tan button. Her notebook says $\sin 50^\circ = 0.766$ and $\cos 50^\circ = 0.643$. Using $\tan \theta = \sin \theta \div \cos \theta$, what is $\tan 50^\circ$ to 3 decimal places?

- A. 0.839
- B. 1.191**
- C. 1.409
- D. 0.492

Correct answer: B. $\tan 50^\circ = \sin 50^\circ \div \cos 50^\circ = 0.766 \div 0.643 \approx 1.191$.

USE AFTER: EXPLAIN PHASE — Explain: Problem C, computing the Nakuru mast height with distance 40 m

Q4. The angle of elevation to the top of a mast is 50° , and the horizontal distance from the observer to the base is 40 m. Using $\tan 50^\circ = \sin 50^\circ \div \cos 50^\circ = 0.766 \div 0.643$ and height = $\tan \theta \times$ horizontal distance, what is the height of the mast to 1 decimal place?

- A. 47.7 m**
- B. 25.7 m
- C. 33.6 m
- D. 30.6 m

Correct answer: A. Height = $40 \times (0.766 \div 0.643) = 40 \times 1.1913 \approx 47.7$ m. Multiplying 40 by sin alone, or by cos alone, or dividing 40 by tan, gives the wrong answers.

USE AFTER: EXPLAIN PHASE — Explain: Problem A, given $\sin \theta = 3/5$, find $\cos \theta$ and $\tan \theta$

Q5. θ is an acute angle in a right-angled triangle and $\sin \theta = 3/5$. Without a calculator, what are $\cos \theta$ and $\tan \theta$?

- A. $\cos \theta = 4/5$ and $\tan \theta = 3/4$**
- B. $\cos \theta = 4/5$ and $\tan \theta = 4/3$
- C. $\cos \theta = 2/5$ and $\tan \theta = 3/2$
- D. $\cos \theta = 3/5$ and $\tan \theta = 1$

Correct answer: A. With opposite 3 and hypotenuse 5, the adjacent side is 4 (3-4-5 triangle), so $\cos \theta = 4/5$. Then $\tan \theta = (3/5) \div (4/5) = 3/4$.

USE AFTER: EXPLAIN PHASE — Explain: Problem B, given $\tan 35^\circ$ and $\cos 35^\circ$, find $\sin 35^\circ$

Q6. For an angle of 35° , $\tan 35^\circ = 0.700$ and $\cos 35^\circ = 0.819$. Using $\tan \theta = \sin \theta \div \cos \theta$, what is $\sin 35^\circ$ to 3 decimal places?

- A. 1.519
- B. 0.855
- C. 0.573**
- D. 1.170

Correct answer: C. Rearranging $\tan \theta = \sin \theta \div \cos \theta$ gives $\sin \theta = \tan \theta \times \cos \theta = 0.700 \times 0.819 \approx 0.573$.

USE AFTER: MODEL BUILDING PHASE — Model Building: extension task on a mast twice as tall at the same distance

Q7. A mast is 95.4 m tall, and an observer stands 40 m from its base. Using $\tan \theta = \text{height} \div \text{distance}$, what is $\tan \theta$ for the angle of elevation θ to the top?

- A. 2.385**
- B. 135.4
- C. 55.4
- D. 0.419

Correct answer: A. $\tan \theta = \text{opposite} \div \text{adjacent} = \text{height} \div \text{distance} = 95.4 \div 40 = 2.385$. Dividing distance by height gives the inverted value 0.419.